Homework World Map and Personal Places

Objective. Create an interesting world map using rnaturalearth, sf, and ggplot2. Color countries according to a variable of your choice. Create a second map add three approximate places from your life.

Part 1 Create a World Map

Load the world data:

```
library(sf)
library(ggplot2)
library(rnaturalearth)
library(rnaturalearthdata)

world <- ne_countries(
  scale = "medium",
  returnclass = "sf"
)

names(world)
```

Choose one variable to color the countries. Possible choices include:

- pop_est: estimated population, gdp_md_est: estimated GDP in millions, continent, subregion, income_grp, Another appropriate column discovered using names(world)

Requirements for the World Map

- Use a meaningful title, include a subtitle explaining the variable, include a legend, use an appropriate color scale.

Part 2 Create another map with Important Places

Create a data frame containing three approximate locations. Use longitude first and latitude second. You may use approximate location.

```
important_places <- data.frame(
  place = c("Born", "Lived", "Went to School"),
  city = c("City 1", "City 2", "City 3"),
  longitude = c(-118.24, -122.42, -121.89),
  latitude = c(34.05, 37.77, 37.34)
)

important_places_sf <- st_as_sf(
  important_places,
  coords = c("longitude", "latitude"),
  crs = 4326
)
```

Add these points to your city map, state map, or country map. Wherever all three appear. I don't care where these places actually are, just pick 3 important or fun or interesting places.

Part 3 Other improvements

Add at least two improvements from the following list:

- Use different colors for the three points.
- Add the city names to the labels.
- Add a line connecting the three locations.
- Use different point shapes for birth, residence, and school.
- Or other fun things to do

Submission Requirements

- Submit an HTML or PDF document containing your R code.
- Include your completed world map.
- Include your map with the three personal locations.
- Include at least two improvements.
- Tell me what your AI prompts were?